

IoT-Based Smart Parking System

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Abstract

The IoT-Based Smart Parking System is designed to help drivers quickly identify available parking spaces using sensor technology and real-time monitoring. The system uses ultrasonic or IR sensors to detect vehicle presence in each parking slot. The sensor data is processed by a microcontroller such as Arduino Uno or NodeMCU and displayed on an LCD screen or mobile application. This solution reduces traffic congestion, saves time, and improves parking management efficiency.

Objectives

- Detect vacant and occupied parking slots automatically.
- Display available parking spaces in real time.
- Reduce traffic congestion caused by searching for parking.
- Enable remote monitoring using IoT technology.
- Improve parking management efficiency.

Components Required

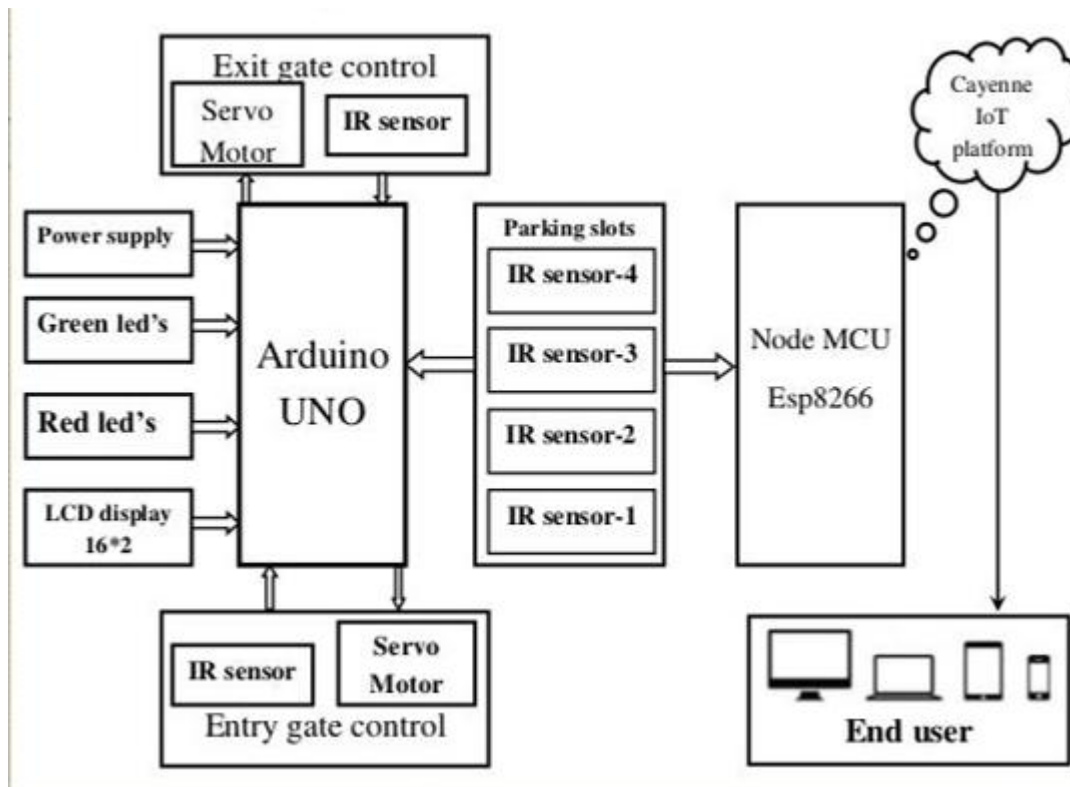
Component	Quantity
Arduino Uno / NodeMCU ESP8266	1
HC-SR04 Ultrasonic Sensors	4
16×2 LCD Display with I2C Module	1
Servo Motor (SG90)	1
LEDs (Red and Green)	4 each
Jumper Wires	As required
Breadboard	1
Power Supply	1

These components are commonly used in beginner-friendly smart parking projects.

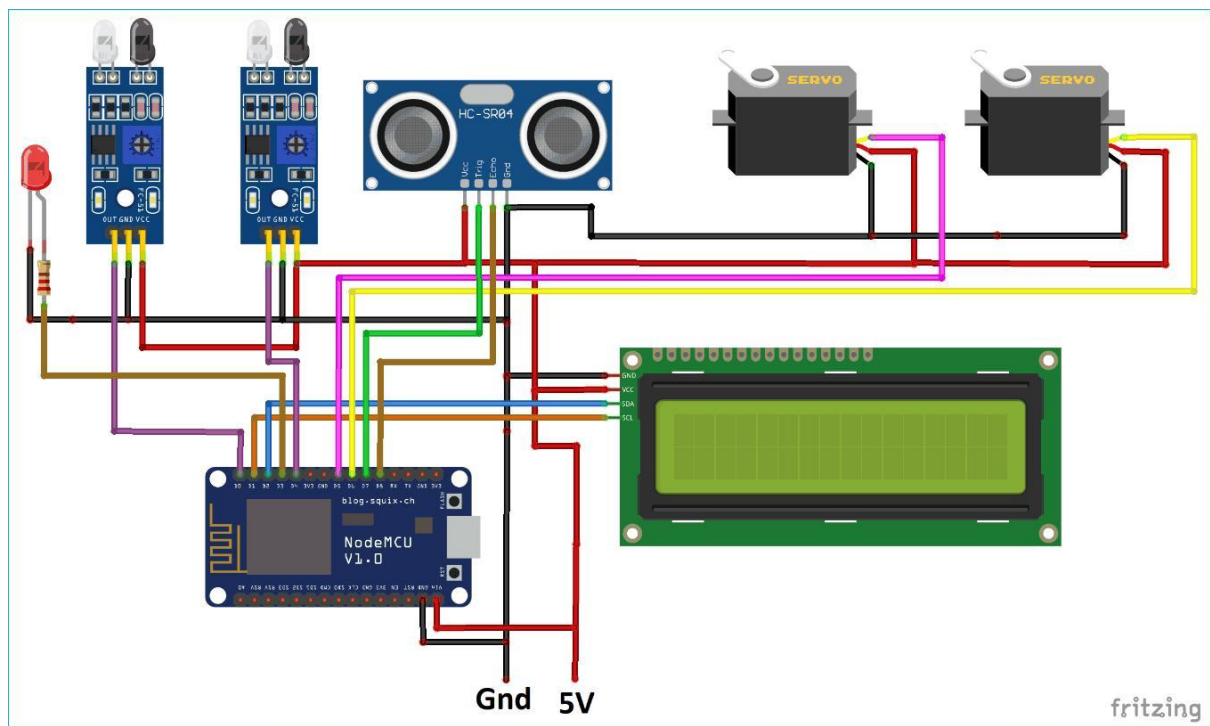
Working Principle

- Ultrasonic sensors are placed in each parking slot.
- The sensors continuously measure the distance to detect whether a vehicle is present.
- The Arduino processes the sensor data and determines slot availability.
- Green LEDs indicate vacant slots, while red LEDs indicate occupied slots.
- The LCD screen displays the number of available spaces.
- If parking slots are available, the servo motor opens the entry gate automatically. Otherwise, the gate remains closed.

Block Diagram



Circuit Diagram



Simple Arduino Code

```
#include <Servo.h>
#include <LiquidCrystal_I2C.h>
Servo gate;
LiquidCrystal_I2C lcd(0x27, 16, 2);
const int trigPin = 9;
const int echoPin = 10;
const int totalSlots = 4;
int availableSlots = totalSlots;
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  gate.attach(6);
  lcd.init();
  lcd.backlight();
  gate.write(0);
  lcd.setCursor(0, 0);
  lcd.print("Smart Parking");
}
void loop() {
  long duration;
  int distance;
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;
  lcd.setCursor(0, 1);
  lcd.print("Slots: ");
  lcd.print(availableSlots);
  lcd.print(" ");
  if (distance < 10 && availableSlots > 0) {
    gate.write(90);
    delay(3000);
    gate.write(0);
  }
  delay(500);
}
```

Future Enhancements

- Mobile application integration using Blynk.
- Cloud data storage and analytics.
- Online parking slot reservation.
- RFID-based vehicle identification.
- AI-based prediction of parking availability.

Conclusion

The IoT-Based Smart Parking System is an effective and low-cost solution for managing parking spaces efficiently. By integrating sensors, microcontrollers, and IoT technology, the system provides real-time parking information, reduces vehicle waiting time, and improves user convenience. It is an excellent mini-project for AIML students to understand the practical applications of IoT and automation.